CommSumm: Community Summaries for Answering Global Questions over Knowledge Graphs

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Abstract

CommSumm addresses Query-focused Summarization for questions that ask about an entire corpus rather than a single passage. It partitions a Knowledge Graph with Leiden Clustering, asks a Large Language Model to pre-summarize each community, and combines the community summaries at query time. This design preserves broad coverage while avoiding the cost of sending every fact to the generator. We introduce GlobalQ-200, a set of 200 global questions judged by human raters for comprehensiveness and diversity. CommSumm wins about 70% of comparisons against GraphHop-style local retrieval. The strongest gains occur when a question asks for themes, contrasts, or recurring entities across distant communities.

Keywords

Global questions, Community summaries, Leiden Clustering, GlobalQ-200

1. Introduction

Local retrieval is effective when a question points to a small evidence set, but global questions ask for coverage across a corpus. GraphHop [1] supplies a useful local-retrieval baseline and graph construction pattern. SubGround [2] similarly targets specific questions through subgraph grounding. CommSumm argues that a corpus-wide question requires a different unit of organization: communities that can be summarized before the final query is known.

2. Method

CommSumm first builds a Knowledge Graph with typed entities and relations, then applies Leiden Clustering to obtain densely connected communities. Each community is represented by its most central entities, relation counts, and source snippets. A Large Language Model writes a short pre-summary for each community with explicit coverage and uncertainty fields. At query time, a query classifier selects relevant communities using their labels and centroid descriptions. The generator combines the selected summaries, resolves overlap, and produces a response organized by theme rather than by source order. A diversity penalty discourages repeated entities, while a coverage check compares requested aspects with the community fields. Pre-summaries are cached, so query cost grows with selected communities rather than with the full graph.

3. Experiments

GlobalQ-200 contains 200 questions evaluated by three human raters. CommSumm is preferred in 69.5% of pairwise comparisons for comprehensiveness and 71.0% for diversity against GraphHop-style local retrieval. Local retrieval produces more precise individual facts, but misses minority themes and cross-community contrasts. Inter-rater agreement is 0.78 on comprehensiveness and 0.74 on diversity.

System	Comprehensiveness wins	Diversity wins
GraphHop-style local retrieval	30.5%	29.0%
CommSumm	69.5%	71.0%

4. Conclusion

CommSumm reframes Query-focused Summarization as coverage over communities. Leiden Clustering and pre-summarization let a Large Language Model answer global questions with broader and more diverse evidence than local retrieval. The main open issue is maintaining factual detail when community summaries compress many related facts.

References

[1] Okafor, L. (2023). GraphHop: Knowledge Graph-Augmented Retrieval for Multi-Hop Question Answering. Conference on Natural Language Systems (CNLS).

[2] Menon, R. (2023). SubGround: Retrieving Query-Specific Subgraphs to Ground Language Model Answers. Conference on Natural Language Systems (CNLS).